

How much productivity is needed to shorten the workweek?

Brazil's 36-hour proposal and 40-hour alternatives

Victor Rangel*

April 2026

Abstract

Brazil's workweek debate began with a 36-hour proposal, but it now includes 40-hour and phased alternatives. This paper computes A_{req} , the one-off TFP gain that would make shorter formal hours output-neutral in a short-run accounting model with informality, firm-size heterogeneity, and an hours-productivity schedule. A 40-hour cap requires a modest productivity gain and leaves the representative agent GHH diagnostic positive. The 36-hour endpoint is more demanding, with an output-neutrality benchmark around 6–8% depending on the hours-efficiency specification and disciplined calibration envelopes of roughly 5.6–7.5% and 6.9–9.2%. The benchmark is large relative to Brazil's recent TFP record, while most of the output cost comes from fewer formal hours rather than from reallocation into informality. The results support evaluating intermediate and phased paths separately from a direct move to 36 hours; they do not rank reforms by social welfare.

Keywords: working hours, informality, total factor productivity, CES aggregator, Brazil, welfare

JEL: E24, J22, J46, O17, O47

*Inspir. E-mail: victorr@insper.edu.br.

1 Introduction

The Brazilian workweek debate now spans more than one statutory endpoint. PEC 8/2025, a proposed constitutional amendment, was introduced as a four-day, 36-hour proposal (Camara dos Deputados, 2025a). By April 2026, the Câmara’s constitutional committee had approved the admissibility of PEC 8/2025 and PEC 221/2019, a special commission had been created, and Câmara coverage reported a relator proposal with 40-hour, 5x2, and phased alternatives (Camara dos Deputados, 2025b, 2026a,b). Much of the debate presumes that shorter hours can be absorbed by productivity gains. The quantitative question is how large that gain would have to be at each point in the hours-cap schedule.

This paper computes the one-off TFP gain that would leave aggregate output unchanged in the short run when the statutory workweek is reduced from 44 hours to alternative caps between 44 and 36 hours. I call this gain the output-neutrality benchmark and denote it A_{req} . The object is a scale benchmark, not a forecast of productivity growth and not a welfare criterion. The exercise asks how large a productivity response would have to be to neutralize the mechanical output loss implied by a shorter cap on normal formal hours. Overtime, compliance, wage bargaining, and changes in total employment are left outside the central counterfactual.

The Brazilian setting complicates the calculation. Informality is 37.8% under the narrow PNAD definition and 44.2% under the broad IBGE one, ranging from 41% in services to 72% in agriculture. Existing models of hours reform tend to abstract from informality (Cacciatore et al., 2016) or treat it through a different margin (Ulyssea, 2018). This paper applies that concern to a broad cut of the statutory cap.

The model is a static partial-equilibrium accounting framework with structural discipline. It combines a CES aggregator over formal and informal effective labor, an hours-productivity function $e(h)$ peaking at 40 hours, two firm-size groups, and GHH preferences for welfare. The substitution elasticity is tied to the Brazilian formal-informal wage premium of Meghir et al. (2015); the resulting interval should be read as a wage-premium-consistent set rather than as a precisely identified elasticity. The central value is an expositional point, while the headline tables report the joint (σ, ω, η_I) envelope.

The central object is A_{req} as a function of the cap. A move from 44 to 40 hours requires a TFP gain of roughly 2%. The 36-hour endpoint is much more demanding: $A_{\text{req}} = 6.63\%$ under the one-sided fatigue specification and 8.18% under the two-sided penalty specification. The joint (σ, ω, η_I) envelope is [5.62%, 7.48%] under one-sided fatigue and [6.93%, 9.23%] under the two-sided penalty. These numbers are large relative to Brazil’s post-1990 TFP record. That comparison is used as a scale benchmark over plausible implementation horizons, not as evidence that any particular reform is infeasible. Aggregate informality rises by +1.6 to 1.9 pp at the 36-hour endpoint, but the main

output-cost channel is the mechanical reduction in formal hours rather than reallocation. The formal-informal margin is therefore more important for incidence than for the headline output-neutrality number. The representative agent GHH calculation is positive for moderate cuts and turns negative below 40h, but it is a calibration diagnostic rather than a social welfare ordering over heterogeneous workers.

Relative to existing hours-reform models, the exercise adds the formal-informal margin and reports the productivity requirement along the hours-cap schedule. It also puts the resulting output-neutrality benchmark on the same scale as Brazil’s historical TFP record.

This paper does not estimate the causal effect of PEC 8/2025, PEC 221/2019, or any specific draft that may emerge from the special commission. It also leaves general equilibrium, wage bargaining, worker selection, price adjustment, and capital deepening outside the central model. The narrower goal is to compute the TFP gain that would render alternative hours caps output-neutral under parameters disciplined by Brazilian data. Evidence from Portugal 1996 (Asai et al., 2024) and Brazil 1988 (Gonzaga et al., 2003) enters as a directional plausibility check rather than as an identified lower bound for Brazil’s effective A_{req} .

The main message is therefore conditional. Within this accounting framework, moderate reductions require much less productivity and can generate welfare gains in the representative agent calibration. The 36-hour endpoint requires a productivity gain that is large by recent Brazilian standards.

Section 2 presents facts, model, and calibration. Section 3 reports calibration checks and external benchmarks. Section 4 reports the counterfactual schedule, the TFP benchmark, and the stress test. Section 5 draws conditional implications for reform design.

2 Brazilian facts and model

Formal hours are concentrated near the statutory maximum. The baseline maps the DIEESE contracted-hours distribution into three model bins: 8.5% of formal workers in the $\leq 39\text{h}$ category are assigned to the 36h bin, 26.9% at 40h remain in the 40h bin, and 64.6% in the $\geq 41\text{h}$ category are assigned to the 44h bin (DIEESE, 2024). PNAD Contínua Q4 2024 habitual hours give a flatter distribution, with an average of 41.2 hours (Instituto Brasileiro de Geografia e Estatística, 2026). Most formal workers operate at or above 40 hours, the margin a 36-hour cap would bind.

Informality is large and dispersed. The national rate is 37.8% under the narrow PNAD definition and 44.2% under the broad IBGE one (Instituto Brasileiro de Geografia e Estatística, 2026). The sectoral range runs from 41.2% in services to 71.8% in agriculture (Table 1).

The formal-informal wage premium is positive and moderate. Meghir et al. (2015) document a conditional wage premium $R \in [1.15, 1.55]$, the ratio of mean formal hourly

wages to mean informal hourly wages among otherwise comparable workers. The calibration below uses this premium to pin down the CES substitution elasticity σ_{sub} .

Brazilian TFP has been stagnant or declining throughout the post-1990 period. The Penn World Table 11.0 `rtfpna` series shows an annualized decline of -0.83% per year from 1990 to 2023 and -0.82% from 1990 to 2019, before COVID (Feenstra et al., 2015; Groningen Growth and Development Centre, 2025; Federal Reserve Bank of St. Louis, 2026). The best rolling post-1990 decade in the updated series is 2000 to 2010, with only $+0.05\%$ per year. The TFP benchmark in Section 4 uses this updated series as the anchor for A_{req} .

These facts call for a model with formal workers concentrated near the cap, substitution between formal and informal effective labor, and per-hour productivity that varies with the workweek. The model adds two firm-size groups for incidence and GHH preferences for welfare.

Table 1: Sectoral and national labor-market moments, PNAD Contínua Q4 2024.

Sector	λ_s	Informality (broad)	Avg. hours (formal)	θ_{36}	θ_{40}	θ_{44}
Agriculture	0.076	71.8%	44.1	0.101	0.269	0.630
Industry	0.205	44.4%	42.5	0.048	0.417	0.535
Services	0.720	41.2%	40.8	0.171	0.410	0.419
National	1.000	44.2%	41.2	0.143	0.406	0.451

Notes: λ_s is the sector’s share of national employment. Informality uses the broad IBGE definition (VD4009 combined with V4017 CNPJ indicator). “Avg. hours” is the mean weekly hours of *formal* workers (V4039, habitual). $\theta_{36}, \theta_{40}, \theta_{44}$ are the shares of formal workers in each contracted-hours bin. *Source:* PNAD Contínua Q4 2024, IBGE microdata, survey-weighted, $N = 209,311$ (Instituto Brasileiro de Geografia e Estatística, 2026).

A representative firm in group $g \in \{\text{Small}, \text{Large}\}$ produces output Y_g from capital K_g and effective labor L_g using a Cobb-Douglas technology. Small firms have at most 49 employees, and large firms have 50 or more.

$$Y_g = A \cdot K_g^\alpha \cdot L_g^{1-\alpha}, \quad (1)$$

where A is aggregate productivity and α is the capital share of income. Effective labor L_g combines formal (F) and informal (I) workers through a CES aggregator,

$$L_g = [\omega L_{F,g}^\rho + (1 - \omega) L_{I,g}^\rho]^{1/\rho}, \quad \rho = (\sigma_{\text{sub}} - 1)/\sigma_{\text{sub}}, \quad (2)$$

where ω is the CES weight on formal labor and σ_{sub} is the elasticity of substitution between formal and informal labor. When $\sigma_{\text{sub}} < 1$ the two inputs are complements, and raising the cost of formal labor does not trigger a large reallocation toward the informal margin.

The two labor inputs are

$$L_{F,g} = N_{F,g} \sum_b \theta_b h_b e(h_b), \quad L_{I,g} = \eta_I N_{I,g} h_I e(h_I),$$

where $N_{F,g}$ and $N_{I,g}$ are the headcounts of formal and informal workers in group g (with $N_{I,g} \equiv N_g - N_{F,g}$ and total group employment N_g fixed), h_b indexes the three contracted-hours bins $b \in \{36, 40, 44\}$ with shares θ_b among formal workers, $h_I = 44$ is weekly hours for informal workers, and η_I is the per-head productivity of an informal worker relative to a formal worker. Per-hour efficiency follows $e(h) = \exp\{-\kappa(h - h^*)^2\}$ above the peak $h^* = 40$, where κ is the curvature of the long-hours fatigue penalty. Below the peak the paper considers two transparent regimes. The preferred *one-sided fatigue* specification sets $e(h) = 1$ for $h \leq h^*$: long workweeks are tiring, but shorter workweeks are not mechanically punished. The conservative *two-sided penalty* specification keeps the same quadratic form below h^* : deviations on either side of 40 hours reduce per-hour efficiency. The one-sided specification is informed by Pencavel (2015), who documents the fatigue leg above the peak but does not characterize efficiency below it. The two-sided specification serves as a conservative upper bound on the required TFP gain.

Each firm chooses the number of formal workers $N_{F,g}$ to maximize profit net of three cost wedges,

$$Y_g - \tau_g N_{F,g} - \frac{1}{2} \pi_m N_{I,g}^2 - \frac{1}{2} \gamma_{F,g} (N_{F,g} - \bar{N}_{F,g})^2, \quad (3)$$

where τ_g is a linear per-worker cost of formality (CLT/FGTS obligations and size-dependent enforcement), π_m is a convex cost of informality (fines, reputational risk, and limited access to formal inputs), and $\gamma_{F,g}$ is a quadratic adjustment cost penalizing deviations from the pre-reform formal headcount $\bar{N}_{F,g}$. The wedge τ_g is calibrated to match group-specific informality and plays the role of a size-dependent distortion in the spirit of Hsieh and Klenow (2009) and Restuccia and Rogerson (2008), with endogenous-formality micro-foundations in Dix-Carneiro et al. (2026). Cutting the statutory cap from $h_0 = 44$ hours to h_1 reduces effective labor through both the hours bins h_b and the efficiency function $e(h_b)$, and widens the wedge between the marginal product of formal labor and its private cost.

The object of interest is the TFP multiplier that restores aggregate output,

$$A_{\text{req}} : \sum_g Y_g(A(1 + A_{\text{req}}), K_g, h_1) = \sum_g Y_g(A, K_g, h_0). \quad (4)$$

A_{req} is the one-off percentage gain in aggregate productivity A that exactly offsets the short-run output loss from moving the cap from h_0 to h_1 . It is an accounting benchmark to compare with Brazil's historical TFP record, not a predicted growth rate.

For welfare, preferences are GHH (Greenwood et al., 1988) with $\sigma_{\text{ghh}} = 1$ and Frisch

elasticity $1/\nu = 0.5$. The compensating variation is

$$\Delta CV = \frac{C_1 - \psi h_1^{1+\nu}/(1+\nu)}{C_0 - \psi h_0^{1+\nu}/(1+\nu)} - 1. \quad (5)$$

The numerator and denominator are the post- and pre-reform consumption-equivalents of welfare under GHH preferences, computed as consumption net of the monetized labor disutility $\psi h^{1+\nu}/(1+\nu)$. ΔCV is the percentage change in welfare.

The model is static and partial-equilibrium. Capital is predetermined, total employment N is fixed, and general-equilibrium feedbacks (wage-price, monetary, fiscal, cross-sector) are absent. Appendix Table A.1 lists the modeled, bracketed, and omitted channels, including compliance, overtime, wage bargaining, labor supply, and sectoral linkages. A textbook user-cost elasticity $\varepsilon_K \in [0.2, 0.4]$ (Caballero, 1999) with capital share $\alpha = 0.35$ implies

$$\Delta A_{\text{req}}^K \approx -\alpha \cdot \varepsilon_K \cdot |\Delta L_F/L| \in [-1.0, -1.8] \text{ pp}, \quad (6)$$

where ΔA_{req}^K is the downward correction to the short-run A_{req} once the endogenous response of the capital stock is factored in. The medium-run analog of the headline falls by 1 to 2 pp over a 3 to 5 year horizon.

Table 2 lists the calibration. External parameters come from PWT, DIEESE, PNAD Contínua, and RAIS (Instituto Brasileiro de Geografia e Estatística, 2026; Ministerio do Trabalho e Emprego, 2024; DIEESE, 2024; Groningen Growth and Development Centre, 2025). The model calibrates τ_g , π_m , κ , and ψ internally. The remaining parameters are set inside ranges consistent with evidence on work hours (Pencavel, 2015; Collewet and Sauermann, 2017), informal productivity (La Porta and Shleifer, 2014), and payroll-compliance rules (Presidencia da República, 1991; Receita Federal do Brasil and eSocial, 2021). Cross-country hours evidence in Bick et al. (2018) supports declining marginal product of hours, but it does not identify a peak. The online appendix reports the source and sensitivity for each parameter.

The CES substitution elasticity σ_{sub} is disciplined by the wage premium rather than directly estimated. From the CES first-order condition, the formal-informal wage premium R is a monotone function of σ_{sub} . Plugging a widened window around Meghir et al.'s estimates, $R \in [1.15, 1.55]$, into the CES FOC at the recalibrated $(\omega, \eta_I) = (0.622, 0.40)$ yields

$$R \in [1.15, 1.55] \implies \sigma_{\text{sub}} \in [1.116, 1.469], \quad (7)$$

$$A_{\text{req}} \in [7.50\%, 8.58\%] \quad (\text{two-sided penalty, central box}).$$

The central value $\sigma = 1.326$ corresponds to $R \approx 1.40$, the midpoint of the MNR window.¹

¹The window $R \in [1.15, 1.55]$ spans the full range of MNR's reported estimates across sub-populations and alternative specifications (Tables 3 and 10). MNR's R is a structural premium estimated inside their search-and-matching model. Inverting it through the CES first-order condition assumes that the relative wage MNR identifies between formal and informal workers coincides with the relative marginal

Table 2: Parameter values by evidential basis.

Symbol	Description	Value	Source
<i>External</i>			
α	Capital share	0.35	PWT + Gollin (2002)
ω	CES formal weight	0.622	1 – narrow informality (Instituto Brasileiro de Geografia e Estatística, 2026)
θ_b	Hours-bin weights (formal)	(0.085, 0.269, 0.646)	Mapped from DIEESE contracted-hours categories (DIEESE, 2024)
ν	Inverse Frisch elasticity	2.0	Standard (Frisch = 0.5)
<i>Moment-matched</i>			
σ_{sub}	CES substitution elasticity	1.326 [1.116, 1.469]	Wage-premium discipline (Meghir et al., 2015); PNAD diagnostic bridge (Instituto Brasileiro de Geografia e Estatística, 2026)
τ_S, τ_L	Formal wedges (small, large)	2.609, 0.004	Bisection to group informality
π_m^S, π_m^L	Informality convex cost	0.000, 42.11	Bisection
κ	Efficiency curvature	2.11×10^{-3}	Derived from e_q at h_{ref}
ψ	GHH disutility weight	5.74×10^{-5}	Baseline FOC
<i>Author-chosen (within ranges consistent with evidence)</i>			
η_I	Informal relative efficiency	0.40	La Porta and Shleifer (2014) informal/formal wage ratio midpoint ($\approx 1/3$ to $1/2$)
e_q	Hours-output elasticity	0.60	Pencavel (2015) range
h^*	Peak-efficiency hours	40	Informed by long-hour evidence in Pencavel (2015)
h_I	Informal hours	44	No statutory cap in informal sector
γ_F^S, γ_F^L	Adjustment costs	0.12, 0.03	RAIS + payroll-compliance rules (Ministerio do Trabalho e Emprego, 2024; Presidencia da República, 1991; Receita Federal do Brasil and eSocial, 2021)

The central value is an expositional reference point, not an identified elasticity. The main quantitative claims therefore report the joint (σ, ω, η_I) envelope alongside the central calibration.

The three wage-related objects play different roles. The formal weight ω fixes the level of formal effective labor using the narrow PNAD informality rate. The informal efficiency η_I fixes the level penalty of informal effective labor, using the broad informal/formal wage-ratio literature. Conditional on those two level parameters, the wage premium R identifies the curvature of formal-informal substitution, σ_{sub} , through the CES first-order condition. The joint (σ, ω, η_I) envelope below varies the three objects independently, so the headline interval does not rely on a single wage moment doing all the work.

This wage-premium-disciplined calibration supersedes earlier author-chosen values of σ_{sub} used in preliminary rounds of the paper. It mainly changes incidence and the sign of the informality response. Under $\sigma_{\text{sub}} < 1$, formal and informal labor are complements, so a tighter formal cap pushes inputs together and mutes the informality response. Under $\sigma_{\text{sub}} > 1$, as disciplined here by the MNR window, they are imperfect substitutes, and the cap pushes marginal formal workers toward informality. For aggregate output, however, the dominant channel remains the mechanical contraction of formal hours: varying σ , ω , and η_I over the disciplined box moves the endpoint by about two percentage points, while the raw hours cut accounts for most of the loss. Letting ω and η_I also vary over their data-consistent ranges ($\omega \in [0.58, 0.66]$, $\eta_I \in [0.33, 0.50]$), the joint (σ, ω, η_I) -box envelope under the two-sided penalty specification is [6.93%, 9.23%] at the eight corners. Under the one-sided fatigue specification the central A_{req} is 6.63% and the envelope is [5.62%, 7.48%]. The online appendix reports the full grids. The small reallocation elasticity documented by Derenoncourt et al. (2025) supports a modest intra-firm σ and is distinct from the high substitution elasticities between worker types reported in refugee studies.

Rebuilding the calibration on PNAD habitual hours ($\theta = (0.143, 0.406, 0.451)$, 44.2% informality) yields $A_{\text{req}} = 7.80\%$ under the two-sided penalty specification. The baseline uses DIEESE contracted hours because the statutory cap is a contracted-hours concept.

product of formal vs. informal effective labor inside the CES aggregator. The online appendix reports a descriptive PNAD/SIDRA bridge using published monthly earnings by formal/informal category pair. The private-employee pair gives $R = 1.24$, while the domestic-worker pair gives $R = 1.58$. These are not controlled microdata regressions, but they show that current published Brazilian earnings gaps are close to the MNR structural window. The mapping from R to σ_{sub} depends on (ω, η_I) . The anchors $\omega = 0.622$ (1 minus the narrow PNAD 2025 informality rate) and $\eta_I = 0.40$ (midpoint of the La Porta-Shleifer informal/formal wage ratio range 1/3 to 1/2) are theoretically consistent. Under competitive wage-taking, the CES aggregator identifies η_I through the marginal-product (wage) ratio rather than through value-added per employee, which is contaminated by capital-per-worker differences. The online appendix documents the derivation.

3 Calibration checks and directional benchmarks

The checks in this section have a narrow role. The model matches the calibration targets by construction. The external benchmarks are directional plausibility checks, not causal validation of the counterfactual. Magnitudes from the Portuguese 1996 reform and the Brazilian 1988 reform should be compared in sign and order of magnitude, not in level.

Panel B1 evaluates the model against domestic moments that are not used to fit the main parameters. The model is close on mean formal hours and on the share of formal workers above 36h. It understates the cross-sector range in average formal hours, which is a limitation of the parsimonious national hours distribution. As a check, the replication package perturbs the formal hours distribution θ within a ± 5 pp envelope around the DIEESE 2024 weights and re-runs the calibration. The fitted Panel B1 moments shift by 0.4 to 0.6 h/week and 1 to 2 pp, with the sign of the gap preserved. These non-targeted moments are informative, but they are not a separate identification exercise.

Panel B2 compares the model with two hours reforms. The Portuguese 1996 reform has a common $44 \rightarrow 40$ h cap, and the Brazilian 1988 reform has a $48 \rightarrow 44$ h cap. Signs match. Aggregate output falls and hourly productivity rises in both data and model. Magnitudes do not match. The firm-level Portuguese estimate of +4.4% in hourly productivity is about 2.6 points above the aggregate model prediction, a gap consistent with worker selection, work intensification, and sectoral capital deepening that the model does not encode. The Brazilian 1988 evidence points in the same direction, with a slightly larger empirical hourly-wage response. These comparisons are directional benchmarks rather than lower bounds for Brazil’s effective A_{req} . Portugal and Brazil 1988 differ from the current Brazilian 36h/40h policy environment in informality, institutions, and reform magnitude.

4 The productivity arithmetic of alternative hours caps

The main lesson is that a 40-hour cap and a 36-hour cap are different economic objects. Moving from 44 to 40 hours cuts the long tail of the contracted-hours distribution and lets many workers move toward the assumed efficiency peak. Moving below 40 hours is no longer mainly about trimming fatigue. It asks the economy to give up hours that the model no longer treats as inefficient. That is why the curve in Figure 1 bends upward rather than sliding down smoothly.

Table 4 reports the 36-hour endpoint, but the useful comparison is the slope of the path. The 40-hour alternative needs only about a 2% one-off productivity gain to restore output. The 36-hour endpoint needs a much larger productivity response: 6.63% under the one-sided fatigue benchmark and 8.18% under the two-sided penalty benchmark. The distinction between these two columns is not a technicality. In the first, shorter workweeks

Table 3: Calibration checks and directional benchmarks.

Moment	Observed	Model	Gap
<i>Panel A: Calibration targets (matched by construction)</i>			
Informality rate, small firms (≤ 49 empl.)	0.500	0.500	0.000
Informality rate, large firms	0.200	0.200	0.000
Formal employment share, small firms	0.590	0.590	0.000
Formal employment share, large firms	0.410	0.410	0.000
Aggregate informality, narrow PNAD	0.378	0.377	0.001
Wage premium R , MNR (2015) central target	1.400	1.400	0.000
<i>Panel B1: Domestic checks outside calibration</i>			
Average weekly formal hours, habitual measure	41.23	42.24	1.01 h/week
Formal share above 36h, habitual measure (%)	85.7	91.5	5.80 pp
Sectoral range in average formal hours	3.31	0.80	−2.51 h/week
<i>Panel B2: External directional benchmarks</i>			
ΔY at 44 $\rightarrow$ 40h cap, Portugal 1996 (firm-level)	−3.20	−2.00	1.20 pp
$\Delta Y/h$ at 44 $\rightarrow$ 40h cap, Portugal 1996 (%)	+4.40	+1.77	2.63 pp
$\Delta W/h$ at 48 $\rightarrow$ 44h, Brazil 1988 (%), GMC 2003)	+7.00	+1.77	5.23 pp

Panel B1 rows 1 and 2 compare PNAD habitual-hours moments with the baseline model moments implied by the contracted-hours calibration. Row 3 reports the range across agriculture, industry, and services in average formal weekly hours. The model value comes from the sectoral extension using the same hours-bin structure, so the negative gap means that the model understates observed sectoral dispersion in formal hours. The aggregate-hours gap is driven mainly by the maintained assumption $h_I = 44$ for informal workers and is treated as a sensitivity margin in the online appendix, not as a validation target (A_{req} shifts by < 0.2 pp for $h_I \in [38, 46]$). Panel B2 compares the model’s aggregate prediction at the 44 $\rightarrow$ 40h cap to the Portuguese firm-level estimate (Asai et al., 2024) and to Brazil’s 1988 reform of comparable cap-cut magnitude (48 $\rightarrow$ 44h, $\sim 8\%$) as documented by Gonzaga et al. (2003). Both external benchmarks suggest the model is conservative on the hourly-productivity response. The model’s $e(h)$ captures fatigue and not worker selection, work intensification, or sectoral capital deepening. Sources: PNAD/SIDRA, RAIS 2022, DIEESE 2024, Meghir et al. (2015), Asai et al. (2024), and Gonzaga et al. (2003) (Instituto Brasileiro de Geografia e Estatística, 2026; Ministerio do Trabalho e Emprego, 2024; DIEESE, 2024).

are not assigned a direct efficiency penalty once hours fall below 40. In the second, moving away from 40 in either direction is costly.

Informality moves in the expected direction, but it is not the villain of the aggregate result. The calibrated economy does shift some marginal jobs toward informality, and the increase matters for incidence. Still, most of the output loss comes from the simpler arithmetic of fewer formal hours. The joint (σ, ω, η_I) envelope reinforces that point: changing formal-informal substitutability, the formal CES weight, and informal relative efficiency moves the endpoint requirement, but it does not turn the 36-hour reform into a small productivity problem. Capital adjustment can shave roughly one to two percentage points from the short-run benchmark over a three-to-five-year horizon, but it does not erase the difference between 40 and 36 hours.

Table 4: Main results at the 36-hour endpoint.

	One-sided fatigue (preferred)	Two-sided penalty (conservative)
<i>Positive predictions (no TFP compensation)</i>		
Output, ΔY	−6.60%	−8.02%
Welfare (GHH), ΔCV	−1.76%	−3.63%
Informality, ΔInf	+1.57 pp	+1.92 pp
Output per hour, $\Delta Y/h$	+2.39%	+0.75%
<i>Benchmark: TFP gain that restores output</i>		
A_{req} , central	6.63%	8.18%
A_{req} , joint envelope	[5.62, 7.48]%	[6.93, 9.23]%

The joint envelope spans the eight corners of the disciplined (σ, ω, η_I) box. The one-sided fatigue envelope lies below the two-sided penalty envelope because the former does not penalize efficiency below 40 hours.

The figure also clarifies what the model is, and is not, saying. The first four hours of reduction are cheap in productivity units because the hours-efficiency schedule is doing helpful work. The next four hours are expensive because the reform is now mostly removing effective labor. Appendix Table A.13 reports the diagnostic checks behind this statement. With no hours-efficiency channel at all, the 36-hour endpoint would require 7.74% rather than 6.63% or 8.18%; shifting the efficiency peak by one hour changes the endpoint more than changing several secondary parameters. This is why I treat the 41–42h welfare maximum below as a model result, not as a free-standing policy rule.

The historical TFP comparison should be read in the same spirit. PWT 11.0 is not a validation exercise; it is a ruler. Brazil’s post-1990 TFP record is weak, so a one-off gain of six to eight percent is a large number on that ruler. The 40-hour alternative lives on a different part of the scale. Appendix Table A.12 translates the one-off requirements into annualized gains over three-, five-, and ten-year phase-ins. That phase-in framing is more useful than asking whether a single historical decade mechanically “matches” the counterfactual.

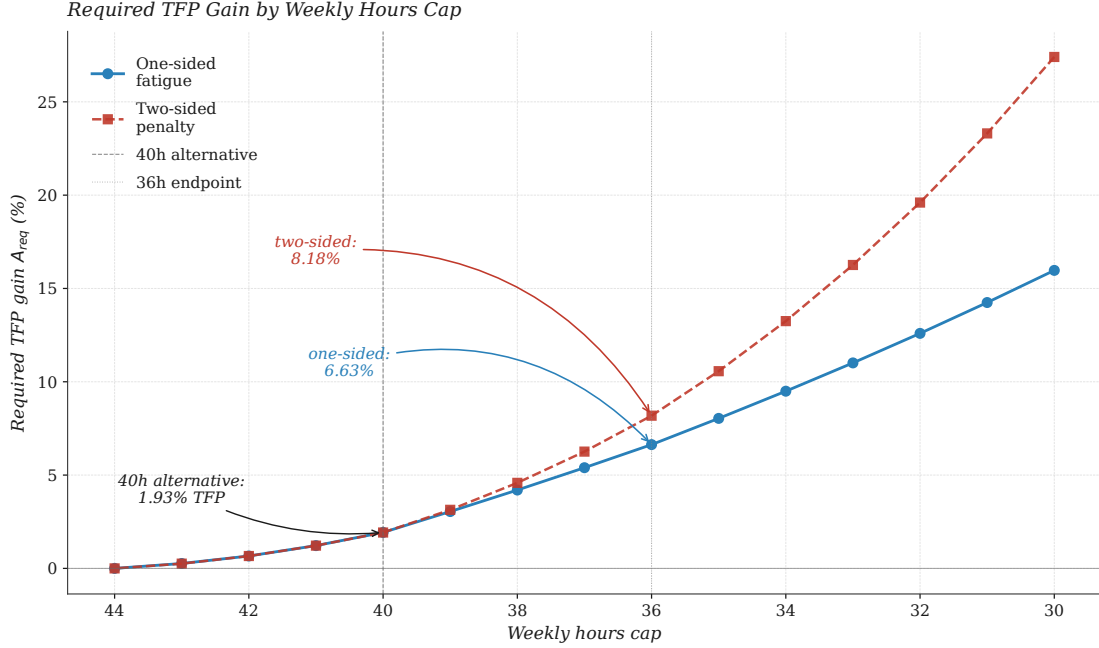


Figure 1: Required TFP gain A_{req} as a function of the weekly hours cap.

Notes: The vertical markers distinguish the 40h alternative in the relator proposal from the 36h endpoint in the original PEC 8/2025 proposal. The one-sided fatigue schedule yields 6.63% at the 36h cap. The two-sided penalty schedule yields 8.18%. The two schedules coincide above $h^* = 40$ and diverge below it, reflecting different assumptions about $e(h)$ below the efficiency peak.

A downside stress test in the appendix asks what happens if the reorganization itself temporarily lowers productivity. This is not the central prediction, but it is a useful guardrail: a one-percent post-reform TFP disruption pushes the two-sided endpoint requirement above nine percent. The reason to report this case is not to forecast disruption. It is to show that the policy arithmetic is sensitive to implementation quality, not only to steady-state labor supply.

4.1 Welfare and incidence

The welfare calculation is best read as a consumption-leisure diagnostic. It asks a narrow question: if employment is fixed, wages do not bargain, and the reform mechanically gives incumbent workers more non-work time, how large is the leisure gain relative to the consumption loss? It is not a social-welfare theorem. Health, household production, bargaining, prices, employment transitions, and distributional weights are outside the calculation.

Within that narrow object, the schedule has a clear shape. Around 41 to 42 hours, the model buys leisure at a relatively low output cost. Below 40 hours, the price of additional leisure rises sharply because the reform is no longer mostly trimming long-hours fatigue. That is why the representative-agent welfare measure turns negative before

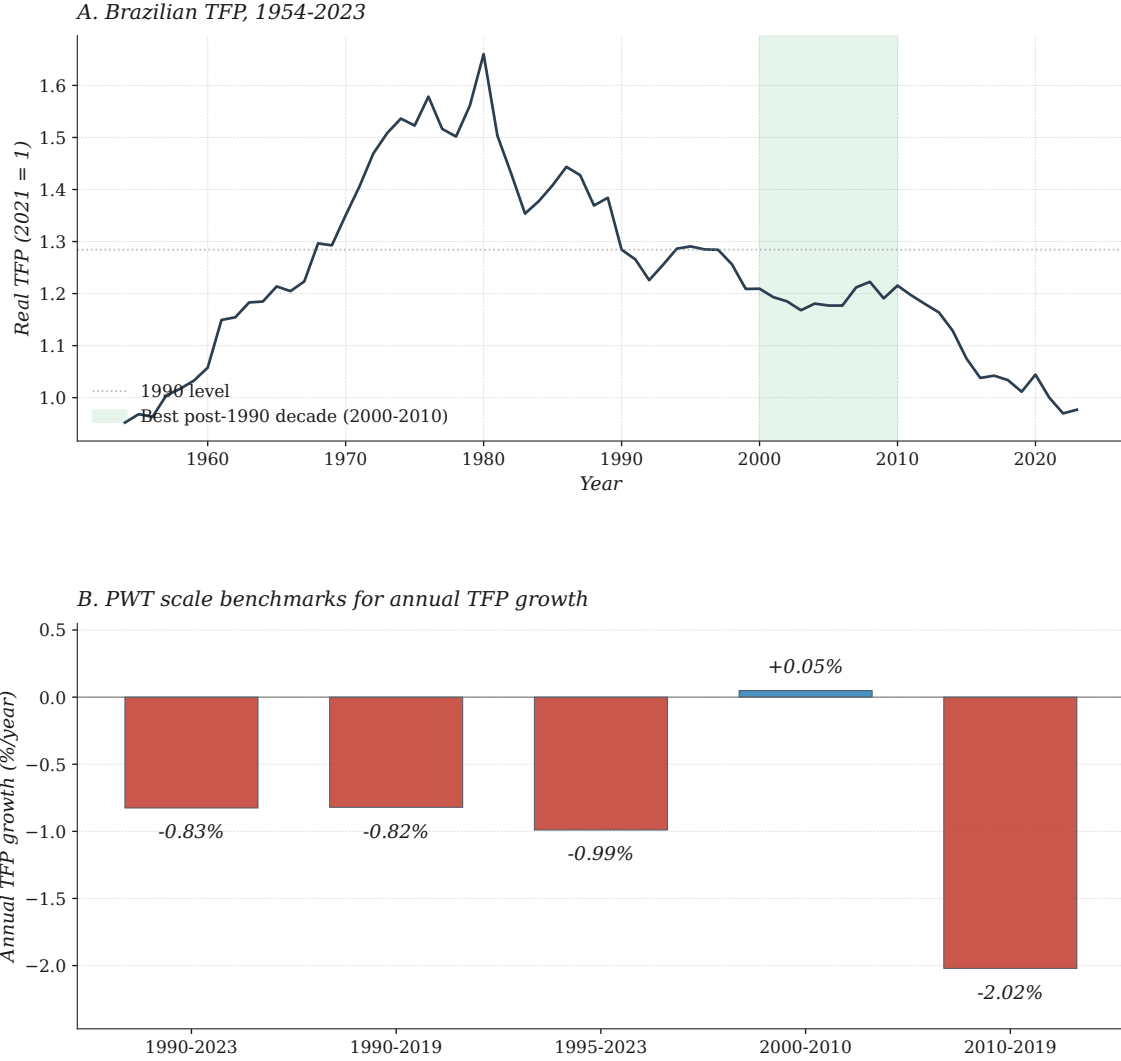


Figure 2: Brazilian TFP and the required one-off productivity gain.

Notes: Panel A plots the PWT 11.0 `rtfpna` series (real TFP at constant national prices) for Brazil, 1954 to 2023. Panel B shows annualized TFP growth rates by horizon. Labels indicate the years required to accumulate the preferred 36-hour baseline $A_{\text{req}} = 6.63\%$ at each historical rate (“never” when the rate is non-positive). Source: Penn World Table 11.0 via FRED series `RTFPNABRA632NRUG` (Feenstra et al., 2015; Groningen Growth and Development Centre, 2025; Federal Reserve Bank of St. Louis, 2026).

the endpoint. The one-sided fatigue and two-sided penalty specifications agree on the moderate-reduction region and disagree mainly once the cap moves below the efficiency peak.

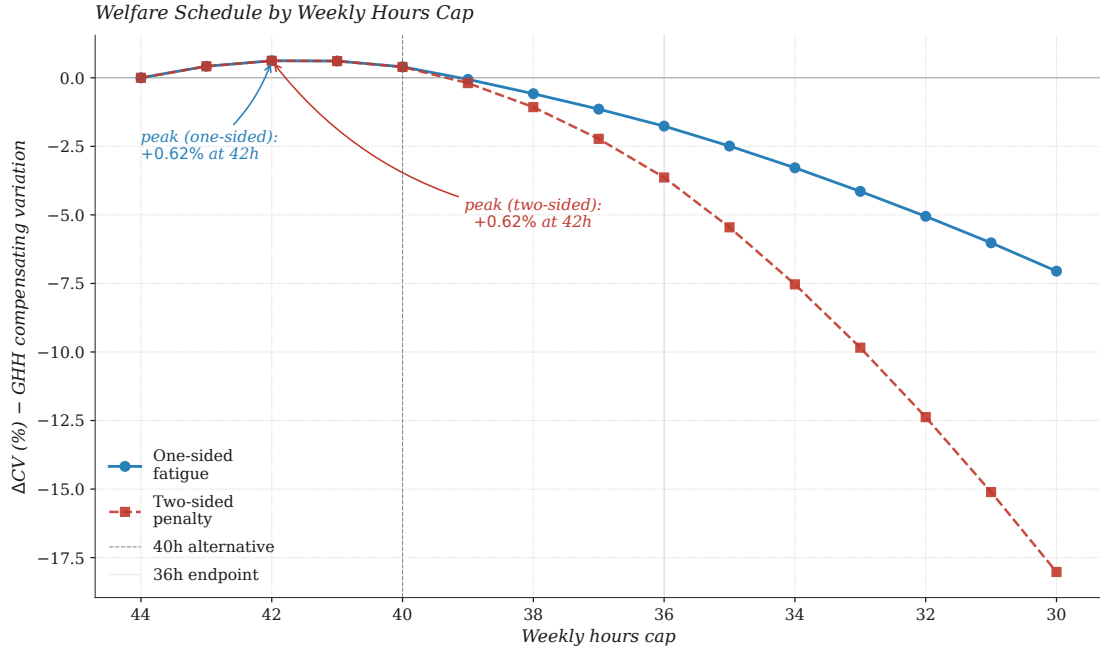


Figure 3: Welfare schedule by hours cap under both calibrations.

Notes: Each line plots aggregate welfare ΔCV from Equation (5) at successively lower hours caps, relative to the 44h baseline, under the one-sided fatigue and two-sided penalty $e(h)$ specifications. ΔCV peaks near 42h at about +0.62% and turns negative below 40h. At the 36h endpoint, welfare falls by 1.76% under the one-sided specification and by 3.63% under the two-sided one.

A TFP gain of approximately 1.4% at the 36h cap is enough to make the endpoint welfare-neutral under the one-sided fatigue specification. The two-sided threshold is 2.9%. Both thresholds lie below the output-neutral A_{req} because leisure partially compensates the consumption loss. Appendix Table A.10 shows that this qualitative ranking survives a separable log and CRRA(2) diagnostic: the 40-hour alternative remains positive, while the 36-hour endpoint remains negative.

The aggregate welfare number masks incidence. In the conservative type-level decomposition reported in Appendix Table A.9, formal stayers gain +1.41% in first-order compensating variation because they work roughly six fewer hours. Workers pushed into informality lose -28.57% . This group is small, 1.9% of workers, but its loss is large. Informal workers not affected by the statutory cap have zero direct incidence in the model. The representative-agent value in Equation (5) differs from the share-weighted type sum because the GHH composite is nonlinear. For this reason, the 41 to 42h welfare maximum should be read as a model-internal representative-agent result, with distributional incidence reported separately.

The welfare schedule does not imply opposition to reductions in working hours. Under

the model’s calibration, the welfare-improving margin lies near 41 to 42 hours rather than at 36. The schedule separates output loss, required productivity, and welfare. These objects move differently with the cap, and the welfare-improving region is narrower than the output-neutral one.

Adjustment costs and informality differ across firm sizes and sectors, producing heterogeneous incidence of the reform. Small firms (≤ 49 employees) require a larger TFP gain than large firms, reflecting their higher formalization wedges and baseline informality. Applying the national framework separately to three broad sectors (agriculture, industry, services) with sector-specific PNAD parameters delivers a per-sector A_{req} of 1.3 to 1.8% for the 40-hour alternative and 6.91 to 8.05% for the 36-hour endpoint under the two-sided penalty specification. Informality rises in every sector at the 36-hour endpoint (agriculture +2.88 pp, industry +2.73 pp, services +2.10 pp). These are sector-by-sector partial-equilibrium counterfactuals, not an input-output incidence analysis. Industry’s A_{req} slightly exceeds agriculture’s because the moderate $\eta_I = 0.40$ combined with industry’s larger formal base dampens the mechanical gap between the two sectors. Services account for most of the aggregate output loss through their output share, not through a larger per-worker effect. The online appendix reports the full decomposition and the parameter sensitivities.

5 Implications for reform design

The relevant policy object is no longer only the original 36-hour version of PEC 8/2025. The live debate also includes 40-hour, 5x2, and phased alternatives. The model therefore reports the full hours-cap schedule. A 40-hour cap has a much smaller output-neutrality benchmark than the endpoint and leaves the representative agent GHH diagnostic positive. A single-step move from 44 to 36 hours is different. It affects most Brazilian formal workers and reduces the labor input of the economy by about 14% mechanically. At that endpoint, the output loss is offset by a one-off productivity gain of 6.63% under one-sided fatigue and 8.18% under the two-sided penalty.

The 36-hour endpoint is large relative to Brazil’s recent productivity record. The PWT comparison should be read as a scale benchmark, because an annual macro residual is not the same object as the short-run productivity response in the model. It is nevertheless informative about magnitudes: post-1990 Brazilian TFP growth has been weak, so the endpoint would require an unusual acceleration if output were to be preserved over a short implementation horizon. Without such a productivity response, the model implies a short-run output loss of 6.6 to 8.0% and a decline in the aggregate GHH diagnostic of 1.8 to 3.6% at the endpoint.

A moderate reduction can raise the model’s GHH diagnostic. In the representative agent calibration, cutting the workweek from 44 to 41 or 42 hours raises that indicator

because the value assigned to extra leisure exceeds the consumption loss. The 40-hour alternative is near the edge of the positive region, and below 40h the calculation turns negative. This result depends on the assumed hours-productivity schedule and representative preferences. It is not a social welfare ordering over heterogeneous workers.

The cost of the endpoint is concentrated. Small firms bear the largest per-worker productivity requirement. Industry and agriculture carry the largest per-sector requirements, and informality rises in all three sectors. Services account for most of the aggregate output loss through their employment and output share, not through a larger per-worker effect. The sectoral exercise is partial-equilibrium and sector-by-sector rather than an input-output incidence model, but it is enough to show that the adjustment burden is uneven.

Informality rises but does not dominate the output cost. The calibrated model, disciplined by the observed formal-informal wage premium, finds an aggregate informality increase of 1.6 to 1.9 pp. That rise matters for welfare accounting, especially for workers pushed into informality. It is small relative to the mechanical hours contraction. The output cost comes primarily from fewer hours worked.

These conclusions have limits. The model is short-run and partial-equilibrium. Endogenous capital adjustment over a three-to-five-year horizon lowers the required productivity gain by about one to two percentage points. Total employment is fixed, which shuts down work-sharing. Evidence on work-sharing from Germany (Hunt, 1996) and Brazil’s 1988 constitutional reform (Gonzaga et al., 2003) suggests that this restriction is compatible with available evidence, but it is still a restriction. Compliance, overtime, organizational disruption, wage bargaining, and informality responses could move the relevant benchmark in either direction. The Portuguese 1996 experience (Asai et al., 2024) documents firm-level productivity gains larger than the model produces, which suggests that the model is conservative on the hourly-productivity response.

Under these limits, the productivity arithmetic suggests that phased paths deserve separate evaluation. An intermediate stop near 40 to 42 hours keeps the productivity requirement far below the 36-hour endpoint and gives capital, informality, and firm-level productivity time to adjust before a further step. The implication is not that the workweek should remain at 44 hours, nor that the model identifies the socially optimal cap. It is that the size and timing of the reduction matter for the tradeoff between output neutrality, representative agent welfare, and incidence across firms and workers.

Online appendix. The public replication repository contains the online appendix with data sources, omitted-channel diagnostics, calibration details, sensitivity grids, sectoral results, welfare incidence, output decomposition, historical TFP table, derivations, and hours-efficiency robustness.

Code and data availability. Code, calibration targets, and unit tests to reproduce every numbered table and figure in this paper are available at the author’s public GitHub repository.

References

- Asai, Kentaro, Marta C. Lopes, and Alessandro Tondini**, “Firm-Level Effects of Reductions in Working Hours,” 2024. Paris School of Economics working paper, November 2024.
- Bick, Alexander, Nicola Fuchs-Schündeln, and David Lagakos**, “How Do Hours Worked Vary with Income? Cross-Country Evidence and Implications,” *American Economic Review*, 2018, *108* (1), 170–199.
- Caballero, Ricardo J.**, “Aggregate Investment,” *Handbook of Macroeconomics*, 1999, *1B*, 813–862.
- Cacciatore, Matteo, Romain Duval, Giuseppe Fiori, and Fabio Ghironi**, “Market Reforms in the Time of Imbalance,” *Journal of Economic Dynamics and Control*, 2016, *72*, 69–93.
- Camara dos Deputados**, “PEC que acaba com a escala de trabalho 6x1 e protocolada na Camara,” 2025. Agencia Camara Noticias, 25 February 2025. <https://www.camara.leg.br/noticias/1136400-pec-que-acaba-com-a-escala-de-trabalho-6x1-e-protocolada-na-camara>.
- , “Relator propoe jornada maxima de 40 horas e restricoes a escala 6x1,” 2025. Agencia Camara Noticias, 3 December 2025. <https://www.camara.leg.br/noticias/1229360-relator-propoe-jornada-maxima-de-40-horas-e>.
- , “Comissao aprova admissibilidade de propostas que acabam com a escala 6x1,” 2026. Agencia Camara Noticias, 22 April 2026. <https://www.camara.leg.br/noticias/1265049-comissao-aprova-admissibilidade-de-propostas-que-acabam-com-a-escala-6x1>.
- , “Motta cria comissao especial para analisar fim da escala 6x1,” 2026. Agencia Camara Noticias, 24 April 2026; updated 27 April 2026. <https://www.camara.leg.br/noticias/1266146-motta-cria-comissao-especial-para-analisar-fim-da-escala-6x1>.
- Collewet, Marion and Jan Sauermann**, “Working Hours and Productivity,” *Labour Economics*, 2017, *47*, 96–106.

- Derenoncourt, Ellora, François Gerard, Lorenzo Lagos, and Claire Montialoux**, “Minimum Wages and Informality,” 2025. NBER Working Paper No. 34445.
- DIEESE**, “Anuário do Sistema Público de Emprego, Trabalho e Renda,” 2024. Departamento Intersindical de Estatística e Estudos Socioeconômicos. Contracted-hours distribution from Table 7 is mapped to model bins as documented in `data_final/calibration_targets.csv`; source checked 2026-04-28. <https://www.dieese.org.br>.
- Dix-Carneiro, Rafael, Pinelopi K. Goldberg, Costas Meghir, and Gabriel Ulyssea**, “Trade and Domestic Distortions: The Case of Informality,” *Econometrica*, 2026, *94* (2), 573–618. Final publication metadata checked against author page and NBER Working Paper No. 28391; DOI not available in the audited records. <https://www.nber.org/papers/w28391>.
- Federal Reserve Bank of St. Louis**, “Total Factor Productivity at Constant National Prices for Brazil [RTFPNABRA632NRUG],” 2026. FRED mirror of Penn World Table 11.0, source ID `rtfpna`; extracted and frozen at `data_raw/fred/RTFPNABRA632NRUG_pwt110.csv`. Checked 2026-04-28. <https://fred.stlouisfed.org/series/RTFPNABRA632NRUG>.
- Feenstra, Robert C., Robert Inklaar, and Marcel P. Timmer**, “The Next Generation of the Penn World Table,” *American Economic Review*, 2015, *105* (10), 3150–3182.
- Gollin, Douglas**, “Getting Income Shares Right,” *Journal of Political Economy*, 2002, *110* (2), 458–474.
- Gonzaga, Gustavo, Naércio Aquino Menezes Filho, and José Márcio Camargo**, “Os Efeitos da Redução da Jornada de Trabalho de 48 para 44 Horas Semanais em 1988,” *Revista Brasileira de Economia*, 2003, *57* (2), 369–400.
- Greenwood, Jeremy, Zvi Hercowitz, and Gregory W. Huffman**, “Investment, Capacity Utilization, and the Real Business Cycle,” *American Economic Review*, 1988, *78* (3), 402–417.
- Groningen Growth and Development Centre**, “Penn World Table version 11.0,” 2025. University of Groningen. Release covers 185 countries through 2023. Checked 2026-04-28. <https://www.rug.nl/ggdc/productivity/pwt/pwt-releases/pwt110>.
- Hsieh, Chang-Tai and Peter J. Klenow**, “Misallocation and Manufacturing TFP in China and India,” *The Quarterly Journal of Economics*, 2009, *124* (4), 1403–1448.
- Hunt, Jennifer**, “Has Work-Sharing Worked in Germany?,” 1996. NBER Working Paper No. 5724; later published in *Quarterly Journal of Economics* (1999).

Instituto Brasileiro de Geografia e Estatística, “Pesquisa Nacional por Amostra de Domicílios Contínua,” 2026. SIDRA tables 4093, 4097, 6374, 6389, and 6413, accessed through <https://sidra.ibge.gov.br> and <https://apisidra.ibge.gov.br>. Frozen inputs stored in `data_intermediate/pnad_targets.json` and `data_raw/sidra/pnadcm_6389_wage_by_position_last.json`; extracted 2026-03-24 and checked 2026-04-28.

La Porta, Rafael and Andrei Shleifer, “Informality and Development,” *Journal of Economic Perspectives*, 2014, 28 (3), 109–126.

Meghir, Costas, Renata Narita, and Jean-Marc Robin, “Wages and Informality in Developing Countries,” *American Economic Review*, 2015, 105 (4), 1509–1546.

Ministerio do Trabalho e Emprego, “Relacao Anual de Informacoes Sociais, ano-base 2022,” 2024. Administrative records of formal employment. Used for formal employment shares by firm size. Public RAIS 2022 files and technical note checked 2026-04-28. <https://www.gov.br/trabalho-e-emprego/pt-br/assuntos/estatisticas-trabalho/rais/rais-2022>.

Pencavel, John, “The Productivity of Working Hours,” *The Economic Journal*, 2015, 125 (589), 2052–2076.

Presidencia da Republica, “Lei No. 8.212, de 24 de julho de 1991,” 1991. Plano de Custeio da Previdencia Social, including employer payroll contributions. Consolidated text checked 2026-04-28. https://www.planalto.gov.br/ccivil_03/leis/18212cons.htm.

Receita Federal do Brasil and eSocial, “Nota Orientativa S-1.0 No. 05/2021,” 2021. Guidance on rural producers’ payroll and commercialization contribution reporting in eSocial. Checked 2026-04-28. <https://www.gov.br/esocial/pt-br/documentacao-tecnica/manuais/nota-orientativa-s-1-0-05-2021-atualizada-em-24-05-2021.pdf/view>.

Restuccia, Diego and Richard Rogerson, “Policy Distortions and Aggregate Productivity with Heterogeneous Establishments,” *Review of Economic Dynamics*, 2008, 11 (4), 707–720.

Ulyssea, Gabriel, “Firms, Informality, and Development: Theory and Evidence from Brazil,” *American Economic Review*, 2018, 108 (8), 2015–2047.